KERAUNOS

Electromagnetic Launch from the Moon, Mars, and Earth

Maglev Space Launch System · SELENE · BRONTE · ASTERIA · ARES

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Perpetual Technologies · Engineering Whitepaper

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0. Executive Summary

Earth, with its thick atmosphere, is the hardest body in the system to launch from electromagnetically. So the program starts on the Moon, where the machine is native, and works outward. The full Earth injector is the long-horizon prize. It is not the opening move.

Launcher	Body	Phase	Character
SELENE	Moon	1 — Proof of concept	No atmosphere: no tube, no plasma window, no aero-shell. Starter track 1–3 km; at 100 g, 2.9 km clears lunar escape in 2.4 s. The first segment is a working exporter, not a demo.
BRONTE	Earth	2–3 — Chimborazo + ASTERIA, in parallel	A 14 km first build on a ~38 km through-mountain line at Chimborazo, Ecuador; crest muzzle at 6,160 m. Proves tube, plasma window, and aero-shell at sub-scale. DoD hypersonic test contracts pay for it.
ASTERIA	Airless bodies	2–3 — (concurrent with BRONTE)	SELENE-class launchers stamped onto Phobos, Deimos, Ceres, the Belt. The core bet — runs alongside Chimborazo, not after it.
ARES	Mars	4 — The port	Mars atmosphere is 0.6% of Earth's; the tube is featherweight. ARES closes the return-trip problem every settlement plan dies on.
BRONTE full	Earth	5 — The endgame	The ~38 km Mach-12 line, then beyond — built only after Chimborazo Phase 1 validates cost and subsystems.

SELENE: Moon (Phase 1)

Parameter	Value	Notes
Track (PoC)	1–3 km, 50–100 g	At 100 g: 1.5 km = orbit speed (1.72 km/s); 2.9 km = escape (2.38 km/s) in 2.4 s
Track (cargo)	10–12 km, 30 g	2.43–2.66 km/s; escape with margin; ~9 s ride
Track (passengers)	~50 km, 3 g	1.72 km/s to low lunar orbit in 58 s (escape-rated 3 g line: ~96 km, future)
Energy per kg	0.41 kWh orbit / 0.78 kWh escape	Orbit ≈ 1/22 of Earth's 8.9 kWh/kg-to-LEO; escape ≈ 1/11
Atmosphere	None	No tube, no plasma window, no aero-shell. The track is the launcher.
Cooling	20 K closed-loop cryoplant	Sealed cryostats hold every coil at 20 K; shadowed cold sinks and ~250 K regolith at 2 m depth help — sunlit surfaces swing far hotter, so the cryostat does the real work

BRONTE + ASTERIA: Earth & Airless Bodies (Phase 2–3 — Chimborazo Project, concurrent with vacuum-world deployment)

Parameter	Value	Notes
Site	Chimborazo, Ecuador	Summit 6,263 m; 1.47°S; massif sits on a 4,000 m plateau

Phase-1 build	14 km, portal 4,069 m	Eastern end of the full line; the only at-grade start the terrain allows
Exit	Crest muzzle 6,160 m, 12.3° east	Terminal bend R = 100 km over final 11 km; ~100 m below peak elevation
Exit air	ρ ≈ 53% SL (p ≈ 46%)	Peak drag and heating roughly halved; weaker ground-level boom
Equatorial bonus	+465 m/s free	Fires east; net launcher requirement to LEO ~7.5 km/s, not ~8
Performance	Mach 8.9 (30 g) / 2.8 (3 g)	Full 38 km line later: Mach 12.6 cargo / 4.6 at human-rated 3 g
Revenue	\$50K–\$500K per test run	Hypersonic test services from day one — the market Auriga Space is already selling into

ARES: Mars (Phase 4)

Parameter	Value	Notes
Atmosphere	~6 mbar (0.6% of Earth)	Featherweight tube; the planet does 99.4% of the plasma window's job
Track (orbit)	~180 km, 3.3 g	Low Mars orbit (~3.4 km/s) in a 105 s run
Track (escape)	~390 km, 3.3 g	Mars escape (~5.0 km/s) in a 155 s run
Energy per kg	1.6 kWh orbit / 3.5 kWh escape	Between Moon and Earth in scale; far easier than Earth in engineering
Staging	Phobos and Deimos	Catch points and depots; SEP tugs fly the interplanetary legs

1. SELENE: The Moon Machine

The Moon has no atmosphere. That sentence deletes every hard engineering problem on the Earth list.

On the Moon these things do not exist: the vacuum tube (the track is already in vacuum), the plasma window (no atmosphere to keep out), the aero-shell, the blast shutters, the transpiration cooling. The track lays on regolith, braced and aligned. That is the system.

The starter track is **1–3 km** — airport-runway length. At 100 g, a 1.5 km run reaches lunar orbit speed (1.72 km/s); a 2.9 km run clears escape (2.38 km/s) in 2.4 seconds. Water, regolith, and metal don't mind the g-load. The very first segment isn't a demo. It is a working exporter.

> **0.78 kWh per kg to escape.** Lunar escape costs roughly one-eleventh of Earth-to-LEO launch energy (0.78 vs 8.9 kWh/kg at 8 km/s); lunar orbit, about one-twenty-second. From the Moon you are already partway to everywhere: a Mars-transfer trajectory from the lunar surface costs ~3.0 km/s total — about **1.25 kWh/kg**, roughly one-seventh of Earth-to-LEO energy, and roughly one-fifteenth the energy of flying the same trip up from Earth's surface.

> **The south pole is the right site.** Shackleton-rim and Connecting-Ridge sites see sun 80–90% of the time; permanently shadowed cold traps sit next door; Earth hangs permanently on the horizon for communications. Rev 2 honesty on two Rev 1 claims: polar *surface* temperatures swing widely (shadow cryo-cold to ~200 K+ in sun), so the 20 K coils live in sealed cryostats regardless — the real thermal gift is vacuum, cold sky, and ~250 K regolith two meters down. And the ice: the famous "600 million tonnes" belongs to *north-pole* crater radar (Mini-SAR 2010). For Shackleton itself the honest statement is bounded — radar and reflectance permit up to ~5–10 wt% ice in wall regolith, with no credible tonnage yet; Chang'e-7 (launch ~Aug 2026) goes to ground-truth exactly this. The architecture survives the uncertainty: regolith, oxygen, and metal exports close the early business even in the lean-ice case.

The physics has been settled for fifty years. Gerard O'Neill's NASA Ames summer studies (1976–77) developed the lunar mass driver; Mass Driver One was demonstrated in 1977. A San José State University engineering study (E. Miller, 2023) sizes a polar machine launching 25.4 kg pellets at 2.4 km/s every 10 seconds on 8.7 MW of solar power, ~1.0 MA per bucket coil — SELENE's starter track is exactly that class of machine. What killed every previous program was the same thing: all government funding, no revenue until a Moon base already existed. Cislunar freight removes that deadlock.

The product is propellant — and mass itself. Polar ice (wherever the surveys land), electrolyzed to LOX/LH2 or processed with imported carbon to methalox, becomes propellant for every vehicle in cislunar space, delivered to L1/L2 depots at roughly 10× less than lifting the same kilogram from Earth. Regolith is radiation shielding; sintered metal is structure. The Moon doesn't need factories to matter — it needs to be a source of cheap mass, and it already is.

SELENE grows by ladder — 1–3 km PoC → 10–12 km / 30 g hardened cargo → ~50 km / 3 g passenger line — each step funded by the previous segment's exports. The SELENE kit is the template for every airless body worth working: Phobos, Deimos, Ceres, the Belt. Design once, stamp copies. ASTERIA is the real bet.
(Companion sheet: SELENE Design Map, KER-SEL-TRK-002 — separate document.)

2. The Physics and the Price

The electricity cost of orbital velocity is $0.53 per kilogram. Everything above that is hardware amortization, not physics.

Relation	What it means
$E = ½v^2$ = 32 MJ/kg	At 8 km/s and $0.06/kWh: **$0.53 of grid electricity per kilogram**. Falcon 9 burns ~$200–300K of propellant for ~17–22.8 t to LEO — propellant alone costs ~20–25× KERAUNOS's total electricity spend per kilogram, on a vehicle that is 90% propellant by mass.
$a = v^2/2L$	Acceleration is set by velocity and length. 1,000 km at 8 km/s is 3.3 g — gentle enough for people. StarTram Gen-1 accepted 30 g on 130 km, cargo-only. On the Moon a 3 g orbital track is ~50 km. **Track length is the unlock.**
Kick stage: Isp 350 s, 20% mass	Δv = 3,434 × ln(1/0.8) ≈ **0.77 km/s** for circularization and trim at apogee. That is the only propellant aboard — ~80% of launched mass is payload, vs ~10% for a rocket.
Bends are forbidden at speed	Curving 20° at 4 km/s needs a ~200 km radius. Every high-speed launcher is a straight line with, at most, a gentle payload-g-limited terminal bend. This single rule shaped the entire Chimborazo route (Section 4).

Comparative launch costs (per kg to LEO)

System	Cost per kg	Economic bottleneck
Space Shuttle	$54,500 (full-program-cost basis)	Hand-built; massive refurbishment every flight
Falcon 9 (2026)	~$3,245 list (expendable basis); ~$630 est. internal	$74M list price (Feb 2026); reusable flights carry ~17–18 t, so real per-kg runs higher
Starship (projected)	$100–$1,000 — **aspirational**	V3 flew once (Flight 12, May 22, 2026; payload deploy succeeded, booster mishap, fleet grounded pending FAA review). No reuse-economics data yet; chemical energy density is the hard floor
KERAUNOS	$20–$250 (target)	$0.53/kg electricity; the rest amortizes track capital — the honest unknown is tunnel+tube cost per km, which Phase 1 exists to measure
KERAUNOS + skyhook	toward $5	Kick propellant → ~2–5% when a rotating tether catches pods (Section 8)

3. Why Now: The Signals, as of June 2026

The field validated itself publicly in the last 24 months. The window to enter at peer level is now.

- **SpaceX wants a lunar mass driver — said out loud, February 2026.** Following the company's lunar pivot (Feb 2, 2026), Musk described a mass driver "shooting AI satellites into deep space" at the Feb 11 all-hands, and reaffirmed it March 22 ("Mass drivers on the Moon!"). Honesty requires the caveat: these are founder statements, not a funded program of record — no contracts or milestones have been disclosed. But the largest launch company on Earth pointing at exactly this machine is both validation and a clock. A partner positioned ahead of that build doesn't compete — it supplies the architecture.
- **China is building EM launch hardware, targeting 2028.** Galactic Energy, CASIC, and the Ziyang municipal government are constructing an electromagnetic launch verification platform to throw a Ceres-2-class rocket to **above Mach 1** before ignition. Progress is real: a full-system ejection test of a 1.4 m rocket model (April 2025) and a matrix-switched segmented-drive trial at ~90% system efficiency (January 2026). It's launch-assist — the rocket still carries the full propellant penalty — but the supply chain and urgency are real.
- **The US military already funds this class of technology.** Auriga Space has raised $12.2M cumulatively (VC plus AFWERX, including a $1.25M Direct-to-Phase-II SBIR) for an EM launch-assist track, selling hypersonic test services as Phase-1 revenue — Prometheus (lab) → Thor (outdoor track, 2026) → Zeus (orbital). General Atomics EMALS (484 MJ flywheel storage, 45 s recharge) has launched aircraft from USS Gerald R. Ford since 2017 through a combat deployment. A 2023 AFOSR-filed report (Peterkin) explicitly recommends evolving EMALS for lunar launch, and a US SBIR program record describes a magnetically levitated sled driven to 7 km/s. The demand signal is public and funded.
- **The boom-shaping toolkit just went supersonic.** NASA's X-59 flew Oct 28, 2025 (67 min), expanded its envelope through spring 2026, and made its **first supersonic flight June 5, 2026** (Mach 1.1), with Mach 1.4 acoustic-validation points next — flying on a camera-vision cockpit the whole way. The quiet-shock shaping ASTRAPE borrows is being flight-proven right now.
- **The gap nobody fills:** every competitor is fighting the atmosphere — China's launch-assist, StarTram's paper tube, Auriga's track. Nobody is building the launcher where the atmosphere doesn't exist. KERAUNOS starts there; ASTERIA is the open lane, and Earth revenue (Chimborazo) funds the climb.

4. The Chimborazo Project: BRONTE's Earth Demonstrator

Rev 2 replaces the sketch with a site study: real terrain, real geology, one straight line through the mountain.

Chimborazo, Ecuador. Summit 6,263 m, 1.47°S. The case still holds: the equator gives +465 m/s of Earth's rotation free (fire east); high exit altitude halves the air (at 6,160 m, density is ~53% of sea level — Rev 1 quoted 45%, which is the *pressure* ratio); equatorial launch reaches every inclination. What changed is the engineering, driven by three facts established in the Rev 2 site study:

- **Terrain (SRTM 30 m):** the mountain is a gentle 40-km shield with a concave western approach — no straight surface route to a summit-area exit exists. The answer is a **through-mountain line**: azimuth 095°, passing 0.5 km north of the summit.
- **Geology (IG-EPN, Bernard et al., Beauval et al.):** the andesite core is TBM-proven rock (Ecuador's own Coca Codo Sinclair bored 24.8 km with 9 m machines); the volcano is dormant-but-potentially-active (last eruption ~550 CE, ~1,000-yr recurrence); the governing hazard is seismic — the Pallatanga fault (1797 Riobamba, M~7.6) sets a ~0.4 g design basis, which drove A-frame piers, radial truss bracing, and re-trimmable track jacks. The route avoids the Riobamba debris-avalanche deposit entirely.
- **The straight-line rule:** at exit speeds the track cannot bend more than a payload-g-limited R = 100 km terminal arc — which buys the final 11 km a climb from 6.5° to a 12.3° exit and lifts the muzzle ~550 m to the ridge crest.

The line, in two builds

	Phase 1 — build first	Full line — the growth target
Length	**14 km** (portal 4,069 m — the only at-grade start the terrain allows)	**~38 km** (west portal ~1,360 m, Chazojuán valleys)
Works	El Arenal viaduct (A-frame piers 110–590 m) + 1.2 km crest bore + truss muzzle	adds 24 km of deep tunnel, cover ≤ ~880 m (Gotthard runs 2,300 m)
Exit	Same crest muzzle for both: **6,160 m**, 12.3° east, ~100 m below peak elevation	
Cargo (30 g)	2.81 km/s · **Mach ~8.9** · 10 s in tube	3.97 km/s · **Mach ~12.6** · 16 s (16 g bend cap)
Human (3 g)	Mach ~2.8 · 31 s	Mach ~4.6 · 50 s (2.2 g in the bend)
Per shot (15 t gross)	16.5 MWh · 12.4 GW peak	32.8 MWh · 17.5 GW peak — trackside storage; grid sees a 30 MW × ~66 min recharge

The tube itself: Ø4.0 m flight bore inside a Ø6.0 m ring-stiffened vacuum vessel inside a Ø12 m TBM bore, driven by a **360° stator ring** (24 REBCO sectors per ring, hoops every 0.6 m, ~63,000 on the full line) that levitates, centers, steers, and propels from all sides — required, not optional, because the bend presses the 15 t vehicle outward at up to 16 g. The payload bay is **Ø3.2 m × 8 m** (Cygnus-module class) inside the 23 m ASTRAPE fairing-sled. Vacuum isolation gates every 500 m; 20 K/80 K cryo headers full-length with cooler stations every 500 m.

> **The honest open issues, on the record:** El Arenal piers reach ~590 m (Millau's record is 343 m); the plasma window is proven only at centimeter scale and the Ø4 m bore costs 2.8× the baseline window power — it gets the first R&D dollar; the whole massif sits inside the Reserva de Producción de Fauna Chimborazo under Ecuador's rights-of-nature constitution — the legal path may be the hardest engineering on the sheet; and Mach-12 exits are loud regardless of shaping — altitude and the empty eastern corridor are the real mitigations, and test recovery overflies the inter-Andean valley like any sounding-rocket range.

Revenue from day one

DoD hypersonic test services: sounding rockets are single-shot, wind tunnels don't hold real Mach 6+ conditions for minutes, and no facility on Earth fires Mach-9-class test articles from 6,160 m on the equator several times a day. At $50K–$500K per run, one run per week covers operations. The demonstrator is not a cost center — it is a product, and every shot also retires risk on the plasma window, the aero-shell, and the cost-per-kilometer number the full injector's economics live or die on.

ANNEX A follows this page: the complete BRONTE Chimborazo Design Map (KER-BRO-TRK-001 REV N) — route map on real terrain, both build profiles at true 1:1 scale, tunnel & tube sections, the 360° drive ring, ASTRAPE in three views, and nine engineering charts. Every figure on it re-derived and audited June 12, 2026. **ANNEX B** then follows: the SELENE South Pole Design Map (KER-SEL-TRK-002 REV D), the lunar companion to Section 1 — real LRO/LOLA topography, the open-vacuum EOS-1 ring section, the raised-track side profile at true 1:1, and its site & system charts. Each large panel in both annexes is laid out on its own page and rotated to read at full size.